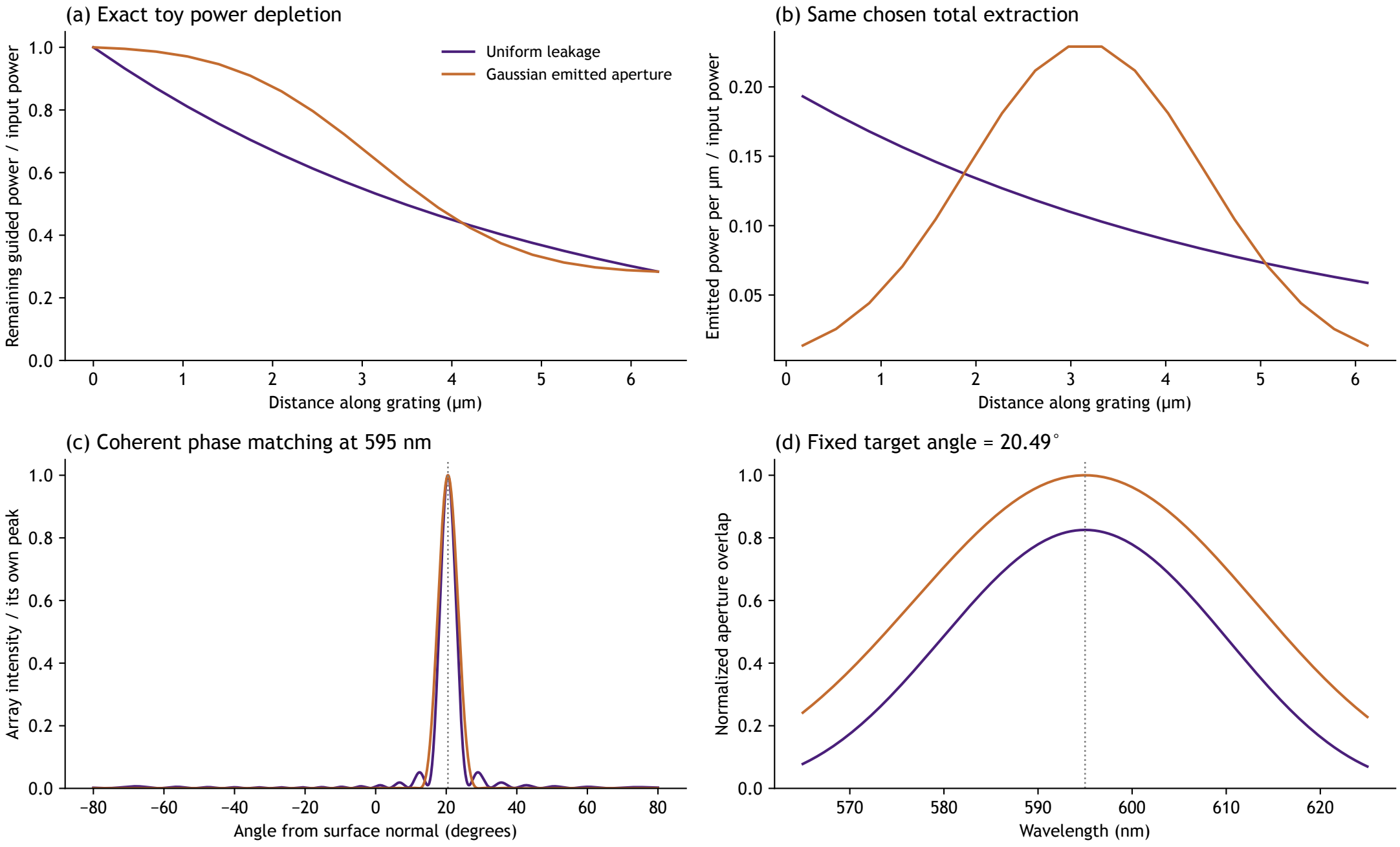
An analytical grating toy: leakage sets the envelope; phase sets the beam
(a) Exact toy power depletion
Uniform leakage
Gaussian emitted aperture
Remaining guided power / input power
Distance along grating (μm)
(b) Same chosen total extraction
Emitted power per μm / input power
Distance along grating (μm)
(c) Coherent phase matching at 595 nm
Array intensity / its own peak
Angle from surface normal (degrees)
(d) Fixed target angle = 20.49°
Normalized aperture overlap
Wavelength (nm)
Illustrative inputs. These are array/aperture calculations, not FDTD fields or predicted fiber efficiencies.